

SCDS TileDB-SOMA Data — New User Guide

A guide to the single-cell data stored as [TileDB-SOMA](#) objects in `s3://rancho-scds-collections/`: what's there, how it's organized, and how to start querying it.

Metrics in this guide were captured on **2026-07-24**. To recompute the current numbers, run `python s3_data_summary.py` (see [Summary metrics](#)).

1. What's in the bucket

Everything lives under `s3://rancho-scds-collections/`, organized into three prefixes:

Prefix	What it holds
<code>tiledb-experiments/</code>	The actual data. One TileDB-SOMA Experiment per dataset (cells × genes: counts, normalized data, cell/gene annotations, embeddings). Grouped into ingestion <code>batch14/ ... batch19/</code> .
<code>collections/</code>	TileDB-SOMA Collections — lightweight groups that <i>reference</i> a curated set of experiments. Four collections split by organism and platform: <code>Human_10x</code> , <code>Human_ss2</code> , <code>Mouse_10x</code> , <code>Mouse_ss2</code> .
<code>manifests/</code>	One Parquet manifest per collection — an aggregated index of every cell's <code>obs</code> metadata across all experiments in that collection, for fast metadata queries without opening each experiment.

How the pieces relate

```

tiledb-experiments/batch14/GSE76312/tiledbsoma_expt  <- physical
Experiment (the data)
    ▲
    | referenced by
collections/Human_ss2/    (a SOMA group: batch14_GSE76312, batch16_E-MTAB-
6678, ...)
    |
    | obs aggregated into
    ▼
manifests/Human_ss2_manifest.parquet  (1 row per cell, all obs columns +
source tags)
  
```

- **Experiments** are the source of truth and hold all expression matrices.
- **Collections** are just groups of references (a few KB each) — opening a collection does not copy data; it points at experiments under `tiledb-experiments/`.
- **Manifests** are a flat metadata cache built from the collections. Use them to explore and filter cell metadata, then open the underlying experiments to pull expression data.

Not every experiment store is wired into a collection: there are **189** experiment stores in [tiledb-experiments/](#) but only **159** are currently included across the four collections. The other **30 stores are not in any collection** and are therefore **not reflected in the metrics below** (which are derived from the collection manifests). These are **intentionally excluded**: they started from **author-provided count matrices** rather than being reprocessed from raw FASTQ through the standard SCDS pipeline, so their upstream processing/QC isn't consistent with the pipeline-processed datasets in the collections. Those 30 hold **~10.0M** cells — almost as many as the collections themselves — including several large datasets:

Excluded store	Cells
batch19/CIMA	5,846,268
batch19/dat-ztf3cc	851,922
batch19/GSE274546	644,098
batch16/LymphoMAP	564,146
batch15/GSE269981	390,756
... 25 more	~1.7M

So the full physical footprint is **~24.3M cells across 189 experiment stores**; the collection metrics below cover the **14.27M cells in the 159 curated experiments**. Bringing an excluded dataset into a collection would generally mean **reprocessing it from FASTQ** through the standard pipeline first (not simply linking the author-counts store), then adding it with [tiledbsoma_collection.py](#) and rebuilding the manifest with [tiledbsoma_manifest.py](#).

2. Summary metrics

As of 2026-07-24. These counts cover only the **159 experiments included in the four collections** (they're computed from the manifests). They exclude the 30 uncurated experiment stores described above (~10.0M additional cells) — see [What's in the bucket](#).

Collection	Experiments	Cells	Genes	Median cells/expt	Max cells/expt
Human_10x	104	11,827,751	36,601	63,482	794,782
Human_ss2	7	14,853	36,522	1,102	7,128
Mouse_10x	46	2,420,528	32,285	34,276	627,495
Mouse_ss2	2	3,414	31,992	1,707	3,050
Total	159	14,266,546	—	—	—

Storage footprint (in [tiledb-experiments/](#), the physical data):

Metric	Value
Total data size	~225 GiB (241.6 GB)
S3 objects	185,228
Experiment stores	189 (batches 14-19)
Manifest files	4 (~1.6 GB total)

Collections themselves are negligible in size (reference groups, a few KB each).

Recompute these numbers

The per-collection counts are derived from the manifests, so they're cheap to refresh:

```
conda activate tiledbsoma
python s3_data_summary.py
```

Storage figures come from the S3 listing:

```
aws s3 ls s3://rancho-scds-collections/tiledb-experiments/ --summarize --
recursive \
    | grep -E "Total (Objects|Size)"
```

3. Getting set up

You need the `tiledbsoma` conda environment (includes `tiledbsoma`, `pandas`, `pyarrow`, `s3fs`, `scanpy`) and AWS credentials with read access to the bucket.

```
conda env create -f tiledbsoma_environment.yml # first time only
conda activate tiledbsoma
```

Quick check that everything works:

```
import tiledbsoma as soma
import pandas as pd

# Read a collection's manifest straight from S3
manifest = pd.read_parquet(
    "s3://rancho-scds-collections/manifests/Human_ss2_manifest.parquet"
)
print(manifest.shape) # (cells, columns)
```

4. Common tasks — where to look

Worked, runnable examples for all of the below are in `tiledsoma_examples_add_manifest.ipynb`:

I want to...	Approach
Explore cell metadata across a whole collection	Load its manifest with <code>pd.read_parquet(...)</code> and use normal pandas (<code>value_counts</code> , filtering, <code>groupby</code>). See the Manifests section.
Open one experiment and inspect <code>obs</code> / <code>var</code> / <code>X</code>	<code>soma.Experiment.open(<expt_uri>)</code> . See the Experiments section.
Query cells by annotation (e.g. cell type) and pull expression	<code>ExperimentAxisQuery</code> with an <code>obs_query</code> value filter → <code>to_anndata()</code> . See Slicing and Filtering .
Find which experiments contain a cell type across a collection	<code>query_cell_type_in_collection(...)</code> in the Collections section.
Find which experiments express a gene	<code>find_experiments_expressing_gene(...)</code> — uses a <code>var_query</code> (gene lives in <code>X</code> , not <code>obs</code>). See the Collections section.
Build/update a manifest for a collection	<code>python tiledbsoma_manifest.py <collection_uri></code> (incremental).

Key identifiers

- **Experiment URI:** `s3://rancho-scds-collections/tiledb-experiments/<batch>/<dataset>/tiledbsoma_expt`
- **Collection URI:** `s3://rancho-scds-collections/collections/<Organism_Platform>`
- **Manifest URI:** `s3://rancho-scds-collections/manifests/<Organism_Platform>_manifest.parquet`
- **Gene identifiers** (in `var`): `var_id` / `gene_symbols` = symbol (e.g. `CD8A`); `gene_ids` = Ensembl ID.
- **Cell-type prediction** (in `obs`): `popv_prediction_ontology_name` (consensus popV call).

5. Repo scripts

Script	Purpose
<code>tiledbsoma_manifest.py</code>	Build/incrementally update a collection's manifest Parquet.
<code>tiledbsoma_collection.py</code>	Assemble collections from experiments.
<code>tiledbsoma_experiment.py</code>	Convert/ingest datasets into TileDB-SOMA experiments.

Script	Purpose
<code>s3_data_summary.py</code>	Print the summary metrics table above from the manifests.
<code>tiledb_soma_examples_add_manifest.ipynb</code>	Runnable examples for all common tasks.